1. Prove that the following function is a solution of the corresponding ordinary differential equation: $y = e^x$, $e^{-x}\frac{dy}{dx} + y^2 - 2ye^x = 1 - e^{2x}$

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2. Find the general solution of each of the following ordinary differential equations: (1) $\frac{dy}{dx} = x(2y+1)$ (2) $\frac{dy}{dx} = \frac{y}{x} + \sqrt{\left(\frac{y}{x}\right)^2 + 1}$ (3) $\frac{dy}{dx} = 3y - 2$ (4) $\frac{dy}{dx} = \frac{3y}{x} + \frac{x+1}{x}$ (5) $\frac{dy}{dx} = \frac{2y}{x+1} + (x+1)^3$

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3. Find the general solution of each of the following ordinary differential equations: (1) $\frac{d^2y}{dx^2} + \frac{2}{1-x}\frac{dy}{dx} = 0$

(2) $2\frac{dy}{dx} \cdot \frac{d^2y}{dx^2} = 1$

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4. Prove that $y = (c_1 + c_2x)e^{-2x} + \frac{1}{8}\sin 2x$ is the general solution of the equation $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = \cos 2x$.
5. Prove that $y = c_1\cos x + c_2\sin x - \frac{1}{3}x\cos 2x + \frac{4}{9}\sin 2x$ is the general solution of the equation $\frac{d^2y}{dx^2} + y = x\cos 2x$.

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6. Find the general solution of each of the following ordinary differential equations: (1) $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 0$

(2) $\frac{d^2y}{dx^2} - 3y = 0$

(3) $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 4y = x$

(4) $\frac{d^2y}{dx^2} + y = 5e^{2x}$